

Exercise 1

For the following function, determine whether there exist continuous extensions at the endpoints of the domain. Then study the differentiability of such extensions.

$$f : [0, \pi/2) \rightarrow \mathbb{R}, f(x) = (\cos x)^{(\pi/2)-x}$$

Setting $y = \pi/2 - x$ we have

$$\begin{aligned} \lim_{x \rightarrow \pi/2^-} f(x) &= \lim_{x \rightarrow \pi/2^-} e^{(\frac{\pi}{2}-x) \log \cos x} = \lim_{y \rightarrow 0^+} e^{y \log \sin y} = \lim_{y \rightarrow 0^+} e^{y \log(y+o(y))} = \\ &= \lim_{y \rightarrow 0^+} e^{y \log(y(1+o(1)))} = \lim_{y \rightarrow 0^+} e^{y \log y + y \log(1+o(1))} = e^0 = 1. \end{aligned}$$

We obtain a continuous extension by setting

$$f(\pi/2) = 1.$$

We examine the limit of the difference quotient:

$$\begin{aligned} \lim_{x \rightarrow \pi/2^-} \frac{f(x) - f(\pi/2)}{x - \pi/2} &= \lim_{x \rightarrow \pi/2^-} \frac{e^{(\frac{\pi}{2}-x) \log \cos x} - 1}{x - \pi/2} = \\ &= \lim_{x \rightarrow \pi/2^-} \frac{(\frac{\pi}{2}-x) \log \cos x}{x - \pi/2} = \lim_{x \rightarrow \pi/2^-} -\log \cos x = +\infty \end{aligned}$$

using the equivalence

$$e^t - 1 \sim t, \quad t \rightarrow 0$$

with

$$t = \left(\frac{\pi}{2} - x\right) \log \cos x, \quad x \rightarrow \pi/2^-.$$

It follows that f can be extended continuously at $x = \pi/2$, but this extension is not differentiable at $x = \pi/2$ (vertical tangent).

Exercise 2

For the following function, determine whether there exist continuous extensions at the endpoints of the domain. Then study the differentiability of such extensions.

$$f : (0, 1) \rightarrow \mathbb{R}, f(x) = (\cos(\pi x/2))^{\log x}$$

We have

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} e^{\log x \log \cos(\pi x/2)} = e^0 = 1$$

since

$$\begin{aligned} \lim_{x \rightarrow 0} \log x \log \cos(\pi x/2) &= \lim_{x \rightarrow 0} \log x \log \left(1 - \frac{\pi^2}{8} x^2 + o(x^2) \right) = \\ &= \lim_{x \rightarrow 0} \left(-\frac{\pi^2}{8} x^2 + o(x^2) \right) \log x = \lim_{x \rightarrow 0} -\frac{\pi^2}{8} x^2 \log x = 0. \end{aligned}$$

We obtain a continuous extension by setting

$$f(0) = 1.$$

We examine the limit of the difference quotient:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} &= \lim_{x \rightarrow 0} \frac{e^{\log x \log \cos(\pi x/2)} - 1}{x} = \\ &= \lim_{x \rightarrow 0} \frac{\log x \log \cos(\pi x/2)}{x} = \lim_{x \rightarrow 0} -\frac{\pi^2}{8} x \log x = 0 \end{aligned}$$

using the equivalences

$$e^t - 1 \sim t, \quad t \rightarrow 0$$

with

$$t = \log x \log \cos(\pi x/2), \quad x \rightarrow 0$$

and

$$\log \cos(\pi x/2) \sim -\frac{\pi^2}{8} x^2, \quad x \rightarrow 0.$$

It follows that f can be extended continuously at $x = 0$, and this extension is differentiable at $x = 0$ with

$$f'(0) = 0.$$

We now examine the behavior for $x \rightarrow 1$ by setting $y = x - 1$:

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} e^{\log x \log \cos(\pi x/2)} = \lim_{y \rightarrow 0} e^{\log(1+y) \log(-\sin(\pi y/2))} = e^0 = 1$$

since

$$\begin{aligned} \lim_{y \rightarrow 0} \log(1+y) \log(-\sin(\pi y/2)) &= \lim_{y \rightarrow 0} y \log(-\pi y/2 + o(y)) = \\ &= \lim_{y \rightarrow 0} y \log(-\pi y/2(1 + o(1))) = \\ &= \lim_{y \rightarrow 0} y \log(-\pi y/2) + \lim_{y \rightarrow 0} y \log(1 + o(1)) = 0 + 0 = 0. \end{aligned}$$

We obtain a continuous extension by setting

$$f(1) = 1.$$

We examine the limit of the difference quotient:

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} &= \lim_{x \rightarrow 1} \frac{e^{\log x \log \cos(\pi x/2)} - 1}{x - 1} = \lim_{y \rightarrow 0} \frac{e^{\log(1+y) \log(-\sin(\pi y/2))} - 1}{y} = \\ &= \lim_{y \rightarrow 0} \frac{\log(1+y) \log(-\sin(\pi y/2))}{y} = \lim_{y \rightarrow 0} \log(-\sin(\pi y/2)) = -\infty \end{aligned}$$

using the equivalences

$$e^t - 1 \sim t, \quad t \rightarrow 0$$

with

$$t = \log(1+y) \log(-\sin(\pi y/2)), \quad y \rightarrow 0$$

and

$$\log(1+y) \sim y, \quad y \rightarrow 0.$$

It follows that f can be extended continuously at $x = 1$, but this extension is not differentiable at $x = 1$ (vertical tangent).

Exercise 3

For the following function, determine whether there exist continuous extensions at the endpoints of the domain. Then study the differentiability of such extensions.

$$f : (0, 1] \rightarrow \mathbb{R}, \quad f(x) = \frac{\sqrt{\cos x} - 1}{x^2}$$

We expand the function $\sqrt{\cos x}$ to order 3 with initial point $x = 0$, using

$$\cos x = 1 - \frac{1}{2}x^2 + o(x^3)$$

and

$$\sqrt{1+y} = 1 + \frac{1}{2}y - \frac{1}{8}y^2 + \frac{1}{16}y^3 + o(y^3)$$

with $y = -(1/2)x^2 + o(x^2)$:

$$\sqrt{\cos x} = 1 - \frac{1}{4}x^2 + o(x^3).$$

It follows that

$$f(x) = \frac{\sqrt{\cos x} - 1}{x^2} = -\frac{1}{4} + o(x)$$

hence f can be extended continuously at $x = 0$ by setting

$$f(0) = -\frac{1}{4}$$

and this extension turns out to be differentiable with

$$f'(0) = 0.$$

Indeed

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} -\frac{1}{4} + o(x) = -\frac{1}{4}$$

and

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} \frac{-\frac{1}{4} + o(x) + \frac{1}{4}}{x} = \lim_{x \rightarrow 0} \frac{o(x)}{x} = 0.$$

Exercise 4

For the following function, determine whether there exist continuous extensions at the endpoints of the domain. Then study the differentiability of such extensions.

$$f : (0, \pi/2] \rightarrow \mathbb{R}, f(x) = (1 - \cos x)^{\log(1+x)}$$

We have

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} e^{\log(1+x) \log(1-\cos x)} = e^0 = 1$$

since

$$\begin{aligned} \lim_{x \rightarrow 0} \log(1+x) \log(1-\cos x) &= \lim_{x \rightarrow 0} x \log\left(\frac{1}{2}x^2 + o(x^2)\right) = \\ &= \lim_{x \rightarrow 0} x \log\left(\frac{1}{2}x^2(1+o(1))\right) = \\ &= \lim_{x \rightarrow 0} x \log\left(\frac{1}{2}x^2\right) + \lim_{x \rightarrow 0} x \log(1+o(1)) = 0 + 0 = 0. \end{aligned}$$

We obtain a continuous extension by setting

$$f(0) = 1.$$

We examine the limit of the difference quotient:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} &= \lim_{x \rightarrow 0} \frac{e^{\log(1+x) \log(1-\cos x)} - 1}{x} = \\ &= \lim_{x \rightarrow 0} \frac{\log(1+x) \log(1-\cos x)}{x} = \lim_{x \rightarrow 0} \log(1-\cos x) = -\infty \end{aligned}$$

using the equivalence

$$e^t - 1 \sim t, \quad t \rightarrow 0$$

with

$$t = \log(1+x) \log(1-\cos x), \quad x \rightarrow 0.$$

It follows that f can be extended continuously at $x = 0$, but this extension is not differentiable at $x = 0$ (vertical tangent).